AI AI &

AI

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1 AI AI &

iPAS AI /

AI AI AI Python

2 (AI & Machine Learning Fundamentals)

(AI) AI

2.1 (Introduction to AI)

· ###

(Artificial Intelligence, AI)
AI

2.1.1 AI

AI

> **Figure Prompt:** Generate a flat illustration comparing three types of AI. 1. Weak AI: A smartphone showing Siri/Assistant icon. 2. Strong AI: A robot shaking hands with a human, looking equal. 3. Super AI: A glowing, abstract digital brain hovering above a city, symbolizing superior intelligence. Style: Modern, clean, vector art.

1. **(Artificial Narrow Intelligence, ANI)** AI
   - AlphaGo Siri Netflix AlphaGo
2. **(Artificial General Intelligence, AGI)** AI
   - (Her) AGI
3. **(Artificial Super Intelligence, ASI)** AI AI

2.2 (Machine Learning Basics)

2.2.1

(Machine Learning, ML) AI AI

-
- …

Figure Prompt: Generate a diagram contrasting "Traditional Programming" vs "Machine Learning". Top flow: Rules + Data -> [Traditional Programming] -> Answers. Bottom flow: Data + Answers -> [Machine Learning] -> Rules. Use icons for Data (document), Rules (scroll), and Answers (lightbulb).

2.2.2 (AI)

1. **(Linear Regression)**
   -
   -
2. **(Logistic Regression)**
   - 0 1
   -
3. **(Decision Tree)**
   - 20 /
   -
   - > 100 -> ->

   Figure Prompt: Generate a visualization of a Decision Tree. The root node asks "Is it raining?". Left branch "Yes" -> "Is it windy?" -> "Don't go out". Right branch "No" -> "Go out". Style: Clean, flowchart style with icons.
4. **(Random Forest)**
   - A A
5. **(SVM)**
   - SVM

6. **K- (KNN)**
    - KNN K
7. **K- (K-Means)**
    - K-Means K

2.3 (Deep Learning & Neural Networks)

2.3.1

(Deep Learning) (Artificial Neural Networks)

- **(Neuron)**
- **(Layer)**

> **Figure Prompt:** Generate a diagram of a Neural Network. Left side: Input Layer (nodes). Middle: Several Hidden Layers (nodes connected by lines). Right side: Output Layer (nodes). Highlight the connections showing data flowing from left to right. Style: Tech, glowing lines, dark background.

2.3.2

1. **(CNN - Convolutional Neural Networks)**
    - CNN
    -
2. **(RNN - Recurrent Neural Networks)**
    - RNN
    -
3. **Transformer**
    - ChatGPT (Self-Attention)
    - (LLM) AI

2.4 (Modeling & Tuning)

2.4.1 AI

AI

1. **(Data Preparation)**
   - **(Garbage In, Garbage Out)**
   - (One-hot Encoding)
   - 180 cm 70 kg
2. **(Training & Evaluation)**
   - 80% 20%
   - **(Cross-validation)**
3. **(Metrics)**
   - **(Accuracy)**
   - **(Recall)**
4. **(Overfitting)** **(Underfitting)**
   -
   -

AI

3 AI AI (AI Applications & Generative AI)

3.1 AI

- **AI AI (L114)**
  - **AI AI (L11401)**
    - AI (Discriminative AI) AI (Generative AI)
    - (/ vs)
  - **AI AI (L11402)**
    -

3.2 AI

- **AI (L12)**
  - **No Code / Low Code (L121)**
    - **No Code / Low Code (L12101)**
      - No Code ( ) Low Code ( )
    - **No Code / Low Code (L12102)**
      - AI
      - Zapier, Make, Coze, Dify
  - **AI (L122)**
    - **AI (L12201)**
      -
      - ChatGPT, Claude, Midjourney, Stable Diffusion, GitHub Copilot
    - **AI (L12202)**
      - Prompt Engineering (Zero-shot, Few-shot, CoT)
  - **AI (L123)**
    - **AI (L12301)**
    - **AI (L12302)**

      - ·
    - **AI (L12303)**

- **AI (L343)**
  - AI

3.3 AI

- **(L21)**
  - **AI (L211)**
    - **(L21101)**
    - **(L21102)**
    - **AI (L21103)**
    - **(L21104)**
    - AI
    -
      - **GAN**
      - **Diffusion Model**
      - **LLM** (Transformer)
  - **AI (L212)**
    - **AI (L21201)**
    - **AI (L21202)**
    - **AI (L21203)**
  - **AI (L213)**
    - **(L21301)**
    - **AI (L21302)**
    - MLOps, Edge AI, Cloud AI, API
- **(L342)**
  - (Bag-of-Words)
  -
  - **NLP** TF-IDF, Word2Vec, BERT, GPT
  - **CV** YOLO ( ), Segmentation ( )

4 (Data Science & Big Data)

4.1

- **(L112)**
  - **(L11201)**
    -
  - **(L11202)**
    -
    - AI
  - **(L11203)**
    -
- **(L331)**
  - (Pie Chart)
  -
  -
    - **(Bar Chart)**
    - **(Line Chart)**
    - **(Histogram)** ( )
    - **(Scatter Plot)**
    - **(Box Plot)**
    - **(Heatmap)**

4.2

- **(L22)**
  - **(L221)**
    - **(L22101)**
    - **(L22102)**
    - **(L22103)**
    -

- **(L222)**
  - **(L22201)**
  - **(L22202)**
  - **(L22203)**
  -
  - Hadoop (MapReduce), Spark ( )
  - **NoSQL** MongoDB ( ), Cassandra ( ), Redis ( )
  - Kafka ( )
- **(L223)**
  - **(L22301)**
  - **(L22302)**
  - **(L22303)**
  -
- **(L224)**
  - **(L22401)**
  - **AI (L22402)**
  - **AI (L22403)**
  - **(L22404)**
  - AI

5 AI (AI Ethics, Governance & Risks)

5.1 AI

- **AI (L11102)**
  - (Privacy) (Security) AI (Ethics)
  - AI
- **(L234)**
  - **(L23401)**
  - **(L23402)**
- **AI (L341)**
  - AI (AI Hallucination)
  - AI
  - **AI**
    - **(Bias)**
    - **Deepfake** AI
    - **(Adversarial Attacks)** AI
    - **(Model Drift)**

5.2 AI

- **AI (L31)**
  - **AI (L311)**
    - (Data Anonymization) (User Consent Management)
    -
  - **AI (L312)**
    -
    -
    - AI

6 AI (AI Programming)

6.1 Python

- **AI Python (L32)**
  - **(L321)**
    -
    - `3_Pig`
    - `print(5, 10, sep=',')` `5,10`
  - **(L322)**
    - `//` ( )
    - `**` ( ), `%` ( )
    -
      - `while...else` `else`
      - `for` (List, String)
      - `break` `continue`
  - **(L323)**
    - List ( ), Tuple ( ), Dictionary ( )
    - Dictionary (key-value) Tuple
    - **(Set)**